Nripesh Niketan

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Summary

Machine learning and systems engineer working across compilers, GPU programming and ML infrastructure. Architect and maintainer of CrossTL, an open-source shader and compute compiler with 1,000-plus GitHub stars and 12 backend targets. 686 merged public pull requests, 96 of them into third-party projects including Apple MLX, PyTorch Lightning, Ray, DeepSpeed and Hugging Face Transformers. Peer-reviewed Springer papers on GPU compiler infrastructure, hybrid game-tree search and LLM tool use. MSc with Distinction; MIET.

Technical Skills

Languages: Python, C++, C, Rust, Go, SQL, Bash, HLSL, GLSL, Metal Shading Language, Mojo
Machine Learning: PyTorch, TensorFlow, JAX, MLX, NumPy, pandas, scikit-learn, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), LLM tool calling and function calling, MMLU evaluation, NNUE, Monte Carlo Tree Search (MCTS), OpenAI API
GPU and Compilers: Nvidia CUDA, AMD HIP, Apple Metal, Metal Performance Shaders Graph (MPSGraph), OpenCL, Vulkan, SPIR-V, DirectX 12, OpenGL, WebGPU (WGSL), Slang, MLIR (Multi-Level Intermediate Representation), LLVM, lexers, recursive-descent parsers, abstract syntax trees (AST), intermediate representation design, code generation, kernel optimisation
Systems and Cloud: Linux, Docker, Git, AWS Lambda, Amazon SQS, Google Cloud Functions, GitHub Actions, CI/CD, pytest, MySQL, MongoDB, multithreading, lock-free concurrency, ARM NEON, SIMD
Data and Automation: Domo, Microsoft Power Automate, Selenium, Tesseract OCR, Oracle ERP, Google Apps Script, ETL pipelines

Experience

Founder, CEO and Principal Engineer, CrossGL LLC
Delaware, USA (Remote from the UK) | May 2024 – Present
- Architected CrossTL, an open-source hub-and-spoke compiler that translates shader and compute code between 12 targets, including Apple Metal, DirectX HLSL, OpenGL GLSL, WebGPU WGSL, Vulkan SPIR-V, Nvidia CUDA, AMD HIP, OpenCL, Rust, Mojo and Slang, through one CrossGL intermediate representation instead of quadratically many pairwise translators.
- Implemented the full pipeline in Python: per-language lexers, recursive-descent parsers, typed ASTs, semantic analysis, IR lowering and target code generation, with bidirectional round-trip translation across seven shader stage families (vertex, fragment, compute, geometry, tessellation, mesh/task, ray tracing).
- Grew the project to 1,015 GitHub stars, 89 forks and 34 contributors as at October 2026; authored 1,089 of its 1,302 commits and 541 merged pull requests, and maintain a 17,500-plus collected-test regression suite with Direct3D 12 WARP and Mesa OpenGL execution gates in CI, plus automated Apache-2.0 releases to PyPI and Zenodo through v3.3.0.
- Built adjacent products: Infrintia, a decentralised GPU compute marketplace with a broker API, host agent and Python SDK; Dyson, a hardware routing library for scientific computing; and Credora, a document intelligence and verification workflow.
- Owned roadmap, releases, code review, contributor onboarding, cloud-credit negotiation and provider partnerships.

AI/ML Engineer Intern, AAICO (The Applied AI Company)
Abu Dhabi, United Arab Emirates | Mar 2024 – Apr 2024
- Built an AI-assisted medical coding workflow in Python using AWS Lambda, Amazon SQS, the OpenAI API, Retrieval-Augmented Generation and vision-based extraction from scanned medical reports.
- Implemented serverless, queue-driven processing to automate report ingestion, classification and structured output generation from unstructured clinical documents.

Developer Relations Lead, Unify
London, United Kingdom (Remote) | Nov 2023 – Apr 2024
- Promoted from Machine Learning Engineer to lead developer relations and contributor operations across Unify's open-source AI projects, managing 85-plus volunteers through recruitment, onboarding, task design, review throughput and recognition.
- Automated the contributor programme with Google Apps Script, Google Cloud Functions, GitHub Actions CI/CD and Vaunt-based contributor recognition.

Machine Learning Engineer, Unify
London, United Kingdom (Remote) | Jun 2023 – Nov 2023
- Contributed 41 merged pull requests to Ivy, a unified ML framework transpiler, covering JAX, PaddlePaddle, TensorFlow and PyTorch frontend compatibility plus Apple Silicon GPU support.
- Wrote PySourceOrganizer, a Python source reformatter for Ivy backend files that normalises import order, class and function layout, assignments and section headers to make large diffs reviewable.

Information Technology Business Analyst Intern, Emerson Electric Co. (now Copeland)
Dubai, United Arab Emirates | Jun 2022 – Jul 2023
- Automated Oracle Advanced Shipping Notice uploads with Python and Microsoft Power Automate, removing roughly two days of manual operations per week.

- Developed Domo ETL pipelines, Oracle data upload scripts, Teamcenter download automation and an OCR-to-Excel tool using Python, pandas, Tesseract OCR and the Domo API.
- Built a multithreaded Selenium scraper for authenticated supplier portals with dynamic navigation, concurrency limits and resilient retry logic.

Publications

Hybrid Chess Search: Combining Alpha-Beta and Monte Carlo Tree Search on Unified Memory Architectures. Niketan, N. *Intelligent Systems and Applications*, Springer Lecture Notes in Networks and Systems, IntelliSys 2026, Amsterdam, pp. 653–664, published online August 2026. doi:10.1007/978-3-032-32726-0_39. NNUE alpha-beta at 7.6M nodes per second on eight threads, GPU Monte Carlo Tree Search through Metal Performance Shaders, and a hybrid of both over lock-free shared state, benchmarked in a 322-game tournament on an M1 Ultra.

Integrating External Tools with Large Language Models (LLMs) to Improve Accuracy. Niketan, N., Santhoshkumar, A., Batatia, H. *Proceedings of ICITA 2024*, Springer Lecture Notes in Networks and Systems, Sydney, pp. 409–421, 2025. doi:10.1007/978-981-96-1758-6_34. The Athena framework reaches 83% on MMLU mathematical reasoning and 88% on scientific reasoning against a 67% and 79% best baseline. 6 citations on Google Scholar as at October 2026, including IEEE ICMI 2026 and Elsevier *Intelligent Systems with Applications*; the extended preprint arXiv:2507.08034 adds comparisons against GPT-4o, LLaMA-Large, Mistral-Large and Phi-Large.

CrossGL: A Hub-and-Spoke Compiler Infrastructure for Universal Shader Translation Across Heterogeneous GPU Architectures. Niketan, N., Shrivastva, V., Srivastav, B., Mishra, S. Accepted at FTC 2026, Berlin, October 2026; Springer Lecture Notes in Networks and Systems, in press.

CrossTL: A Universal Programming Language Translator with Unified Intermediate Representation. Niketan, N., Shrivastva, V. Preprint, arXiv:2508.21256 (cs.PL), 2025.

CrossTL: Universal Programming Language and Translator (software). Niketan, N., Shrivastva, V. Zenodo, 2026. doi:10.5281/zenodo.15826974.

Open Source

Merged contributions to major ML and systems projects: Apple MLX, Ivy, PyTorch Lightning, Ray, DeepSpeed, Hugging Face Transformers and PEFT, Kornia, PyTorch Geometric, Haystack, spaCy Thinc, OpenMMLab MMPreTrain, PyTorch Sparse, Raster Vision, Meta NeuralCompression and hashcat. 686 merged public pull requests in total as at October 2026, of which 96 are in third-party repositories (github.com/NripeshN).

Apple MLX: contributed the softsign, softmax, hardswish and log-softmax activations, the `dropout3d` layer, logical operators, `clip_grad_norm` with `tree_reduce`, `cross` and the orthogonal initialiser across six feature pull requests, plus formatter and pre-commit maintenance across `mlx`, `mlx-data` and `mlx-examples`: 30 merged pull requests, 12th of 344 contributors by commits, first-listed individual contributor in MLX's acknowledgements.

Maintainership and tooling: maintainer of CrossGL/crosstl, Apache-2.0 with 1,000-plus stars and 34 contributors; author of `clang-format-metal`, a pre-commit formatter for C++ and Metal shader sources, and of the upstream change that added Metal file-type support to `pre-commit/mirrors-clang-format`.

Projects

MetalFish – GPU-accelerated chess engine in C++ for Apple Silicon and Nvidia GPUs; the artefact behind the IntelliSys 2026 paper. Runs CPU alpha-beta with NNUE and a GPU Monte Carlo Tree Search driven by an Lc0 BT4 transformer concurrently on one position, exchanging root signals through lock-free shared state. Backends: Metal and MPSGraph, CUDA, and a portable CPU fallback with hand-written ARM NEON and dot-product NNUE kernels. Raised 2-thread MCTS throughput 74-fold by replacing `std::mutex` with `os_unfair_lock`, halving node size and reusing preallocated history buffers. Prebuilt binaries for macOS arm64, Linux x86-64/arm64 and Windows x86-64, with an in-house SPRT test harness.

Panthon – Python cybersecurity attack-simulation library for testing and hardening system defences; published on PyPI across 22 releases.

Education

M.Sc. Cybersecurity and Artificial Intelligence, Distinction, University of Sheffield
Sheffield, United Kingdom | Sep 2024 – Sep 2025
B.Sc. (Hons) Computer Science, Artificial Intelligence, Heriot-Watt University
Dubai, United Arab Emirates | Sep 2021 – Apr 2024
Ph.D. Computer Science (unconditional offer, start October 2027), University of Leeds
Leeds, United Kingdom | Oct 2027 – Sep 2030

Certifications

Member of the Institution of Engineering and Technology (MIET), 2026 | Google IT Automation with Python Professional Certificate | Google Data Analytics Professional Certificate

Awards

50% Heriot-Watt University scholarship with advanced standing
Leadership: President and Captain, Heriot-Watt University Dubai Football Club, Sep 2022 – Jun 2023